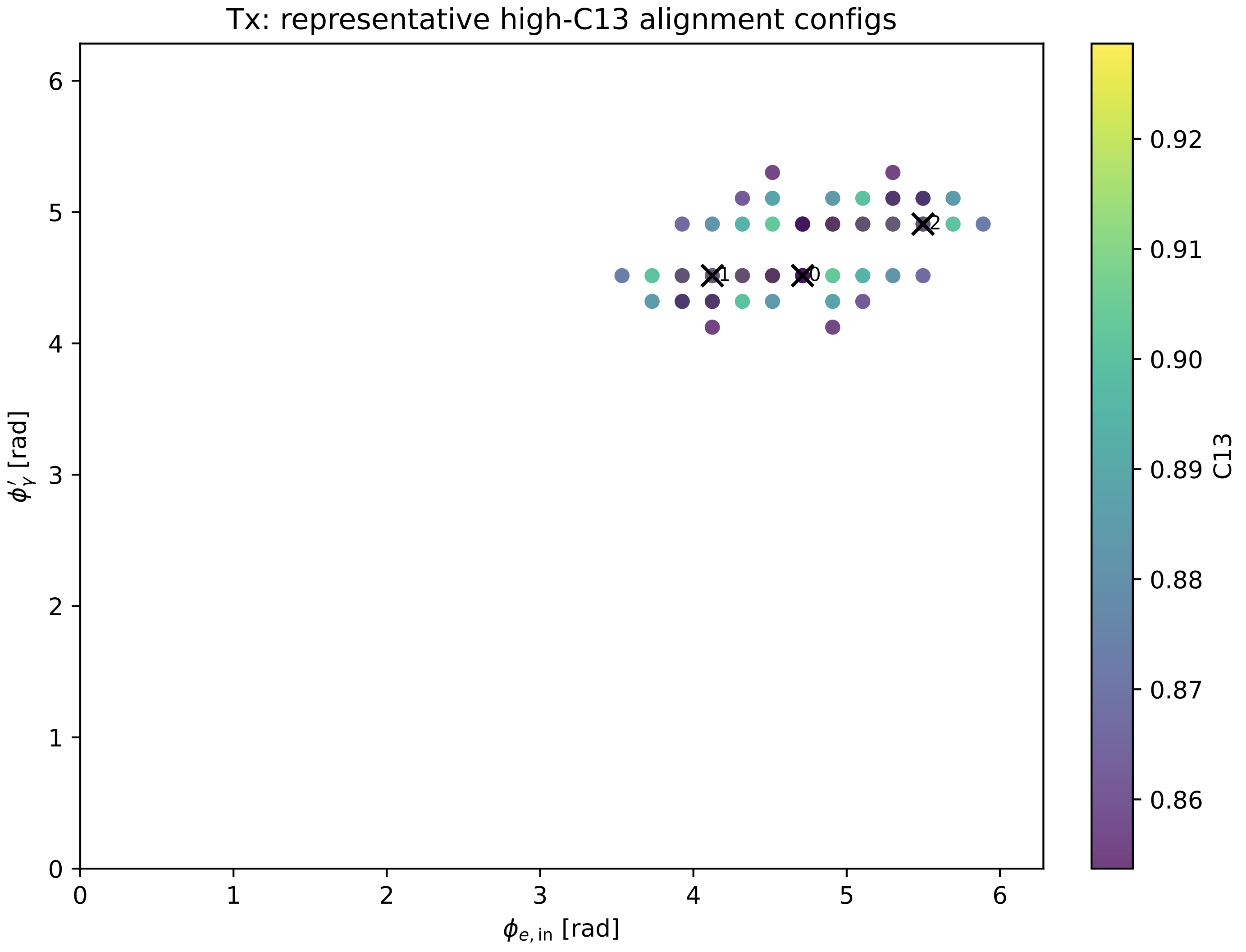
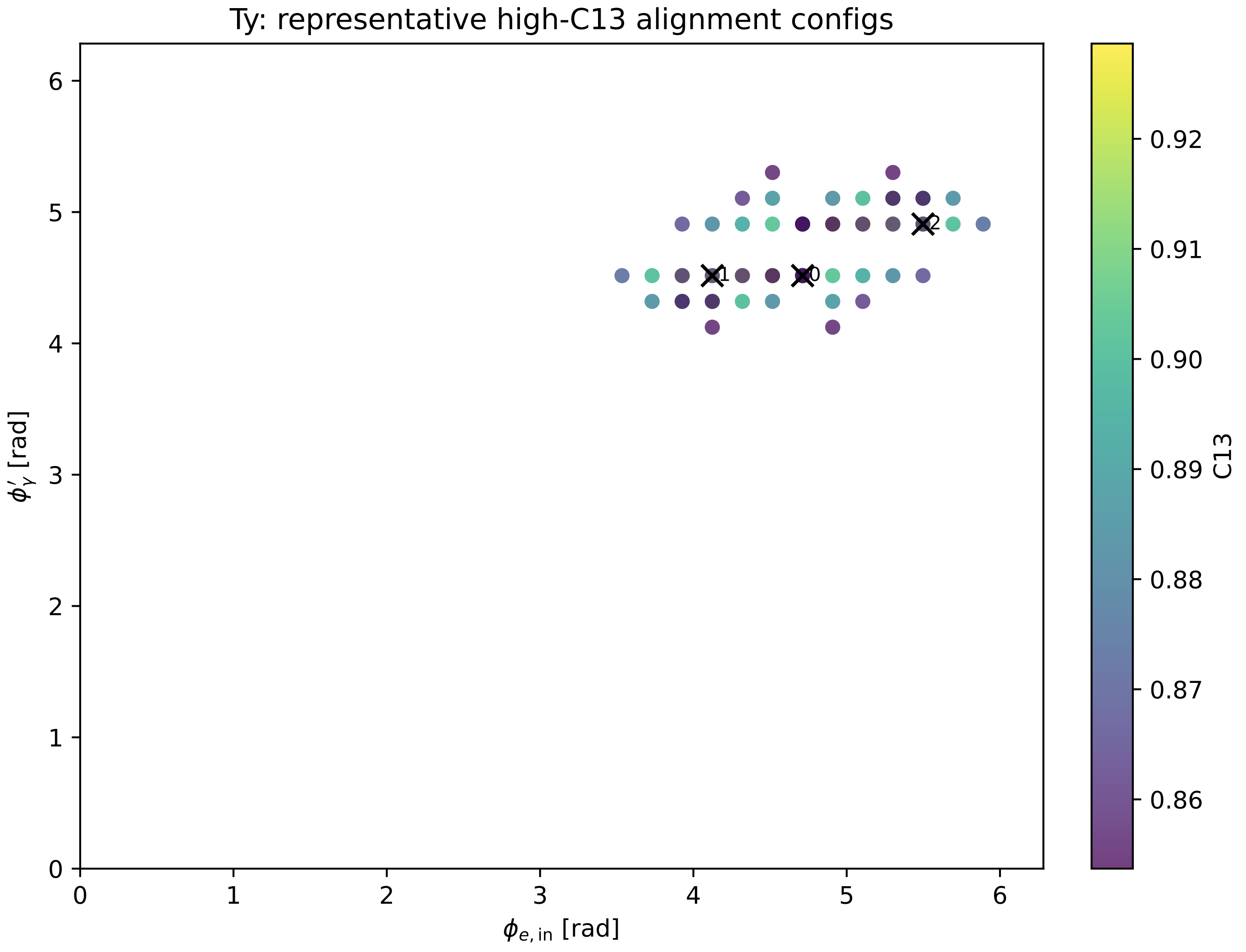


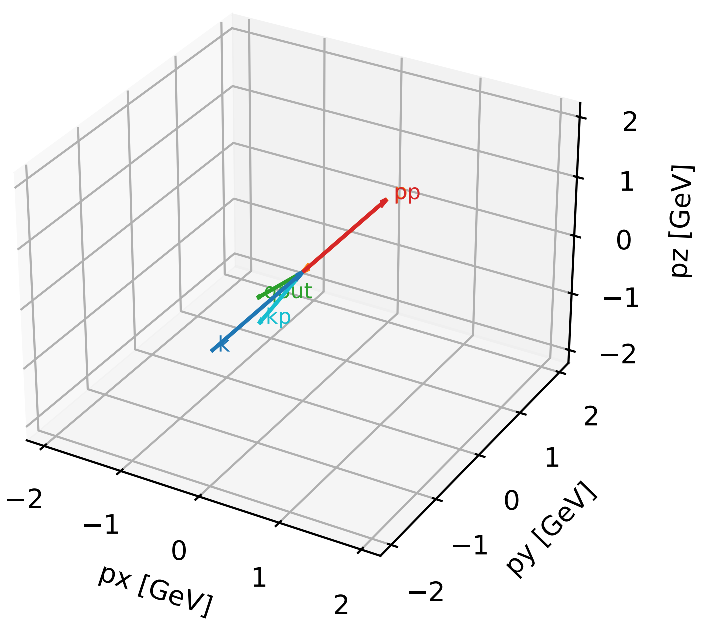
Tx: representative high-C13 alignment configs
ϕ'_γ [rad]
$\phi_{e,\mathrm{in}}$ [rad]
0
1
2
0
1
2
3
4
5
6
C13
0.86
0.87
0.88
0.89
0.90
0.91
0.92



Ty: representative high-C13 alignment configs
ϕ'_γ [rad]
$\phi_{e,\mathrm{in}}$ [rad]
C13
0
1
2
0.92
0.91
0.90
0.89
0.88
0.87
0.86

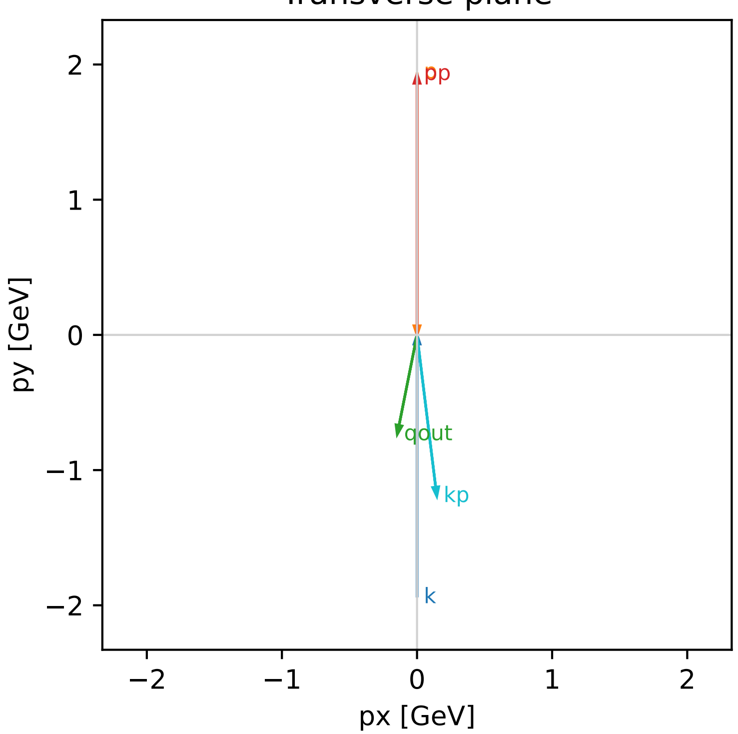
Tx characteristic high-C13 configuration

3D momenta

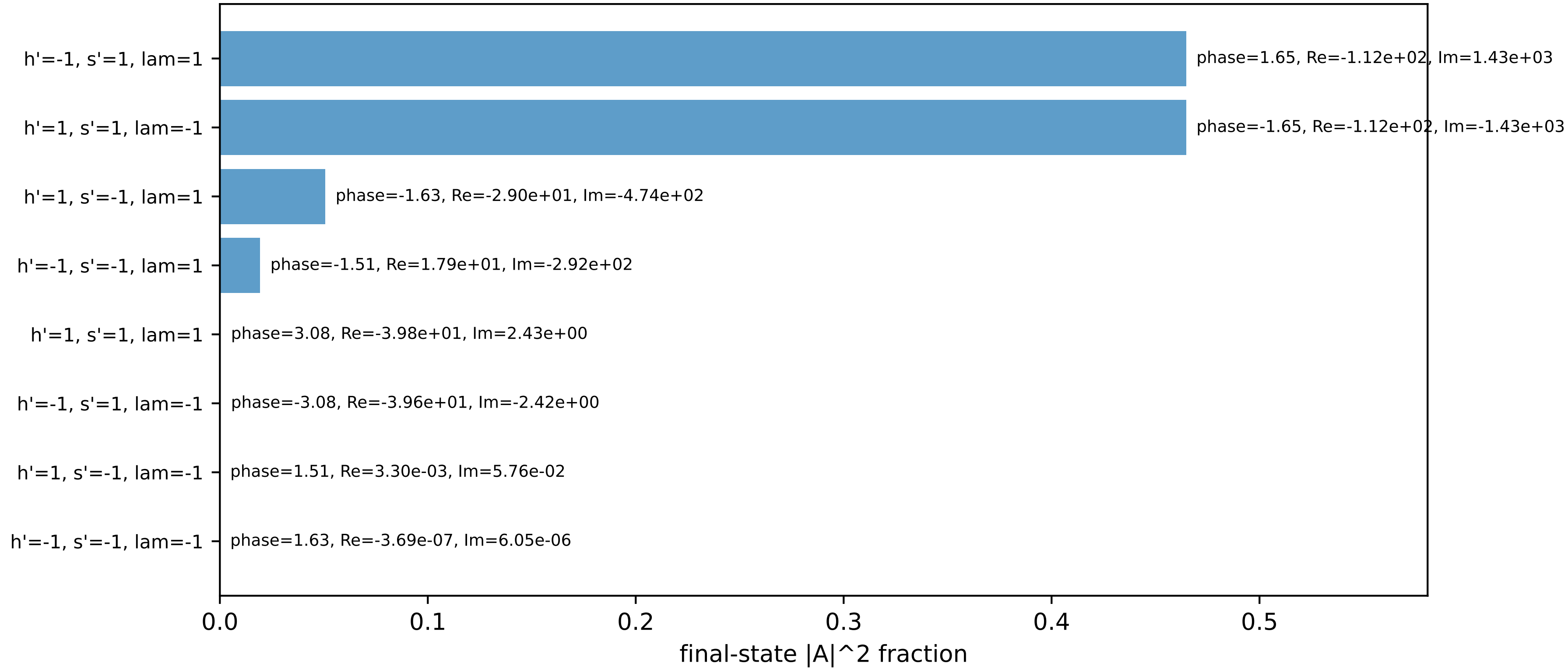



```
Tx_cluster_0  C13=0.928651
kinematic point: low_s_high_theta_in_low_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)

s=16.7824, sqrt(s)=4.09663, pIn=1.94093, pOut=1.92864
theta_in=1.57079, phi_e_in=4.71239, phi_p_in=1.5708
qOut=0.75, phi_gamma=4.51604
Q2=0.0346994, xB=0.00569867, t=-2.87478e-05, W2=6.93417, y=0.382897
theta(e',gamma)=18.242 deg

four-momenta [E, px, py, pz] GeV:
k     E=  1.9409 px= -0.0000 py= -1.9409 pz= -0.0000
p     E=  2.1557 px=  0.0000 py=  1.9409 pz=  0.0000
kp    E=  1.2020 px=  0.1463 py= -1.1930 pz=  0.0000
pp    E=  2.1446 px=  0.0000 py=  1.9286 pz=  0.0000
qout  E=  0.7500 px= -0.1463 py= -0.7356 pz=  0.0000
```

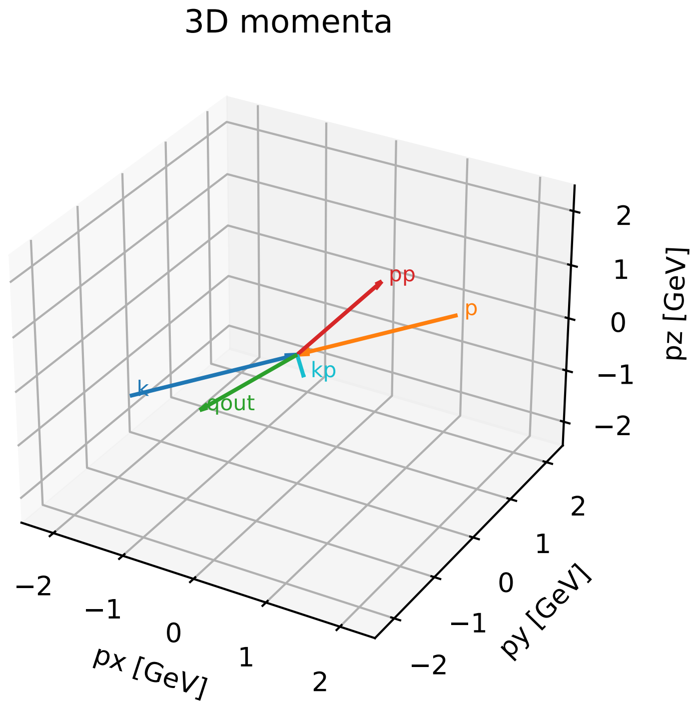
Transverse plane



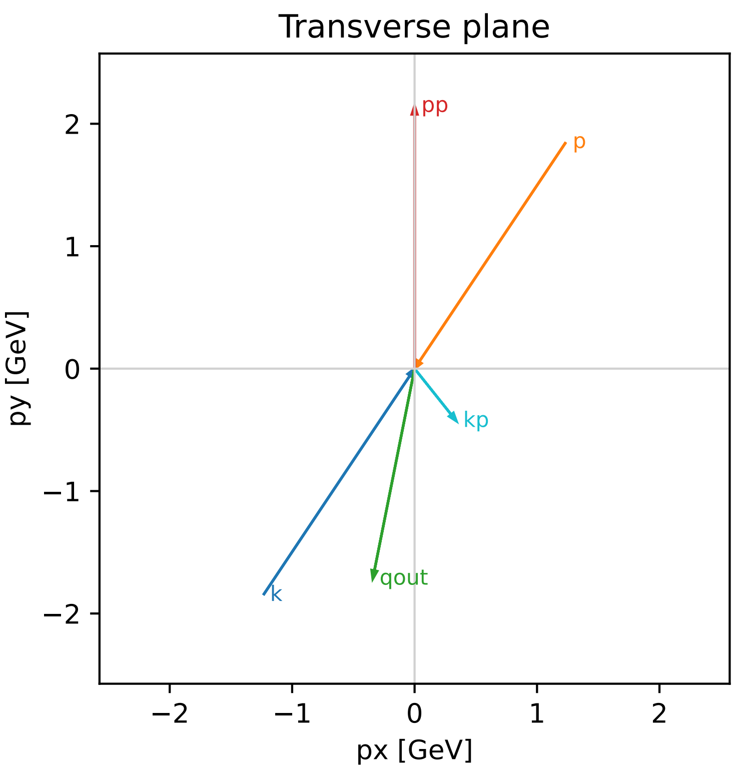
Final-state amplitude decomposition

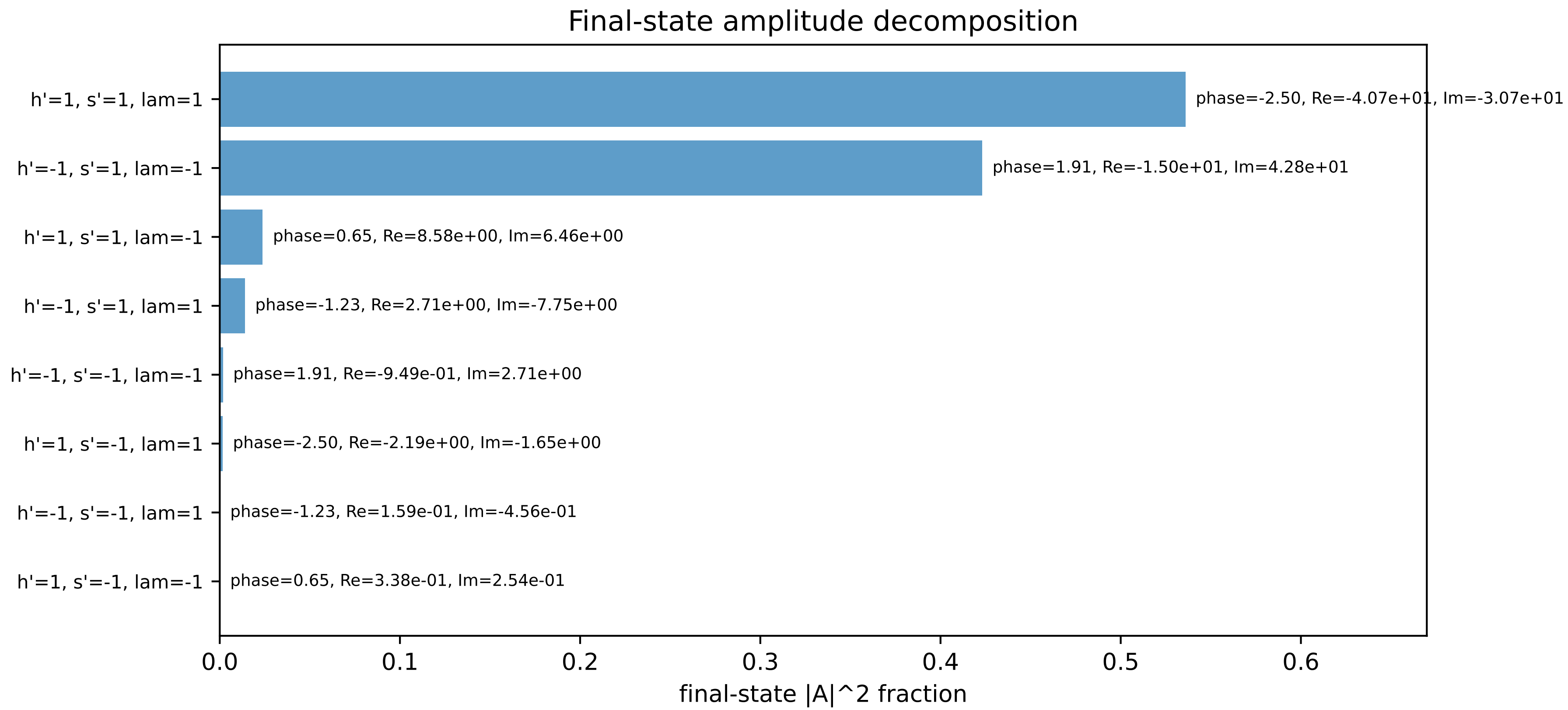


Tx characteristic high-C13 configuration

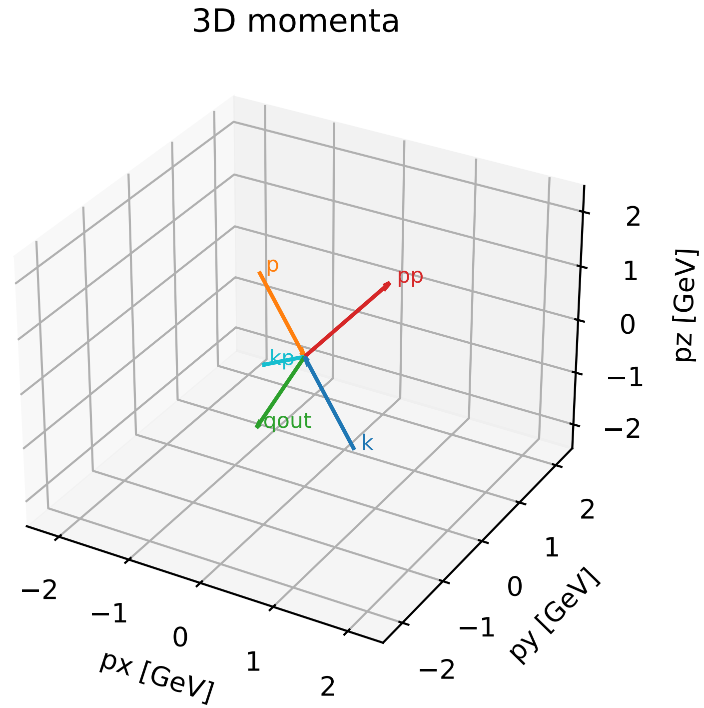



```
Tx_cluster_1  C13=0.919126
kinematic point: mid_s_high_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)

s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=2.14466
theta_in=1.57079, phi_e_in=4.12334, phi_p_in=0.981748
qOut=1.75, phi_gamma=4.51604
Q2=1.69626, xB=0.0983273, t=-1.60898, W2=16.4347, y=0.835974
theta(e',gamma)=49.8104 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.2244 px= -1.2358 py= -1.8495 pz= -0.0000
p     E=  2.4141 px=  1.2358 py=  1.8495 pz=  0.0000
kp    E=  0.5477 px=  0.3414 py= -0.4283 pz=  0.0000
pp    E=  2.3408 px=  0.0000 py=  2.1447 pz=  0.0000
qout  E=  1.7500 px= -0.3414 py= -1.7164 pz=  0.0000
```

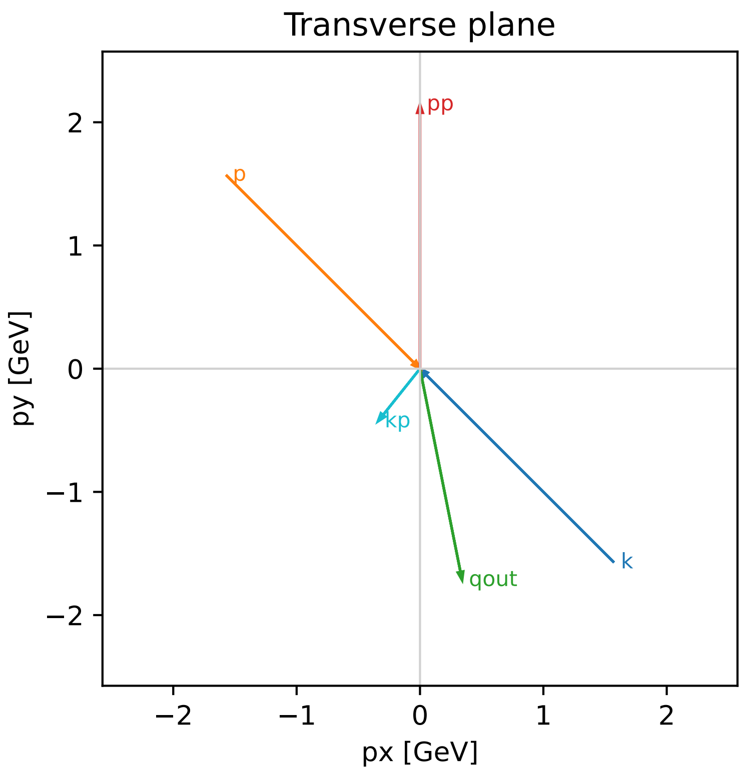


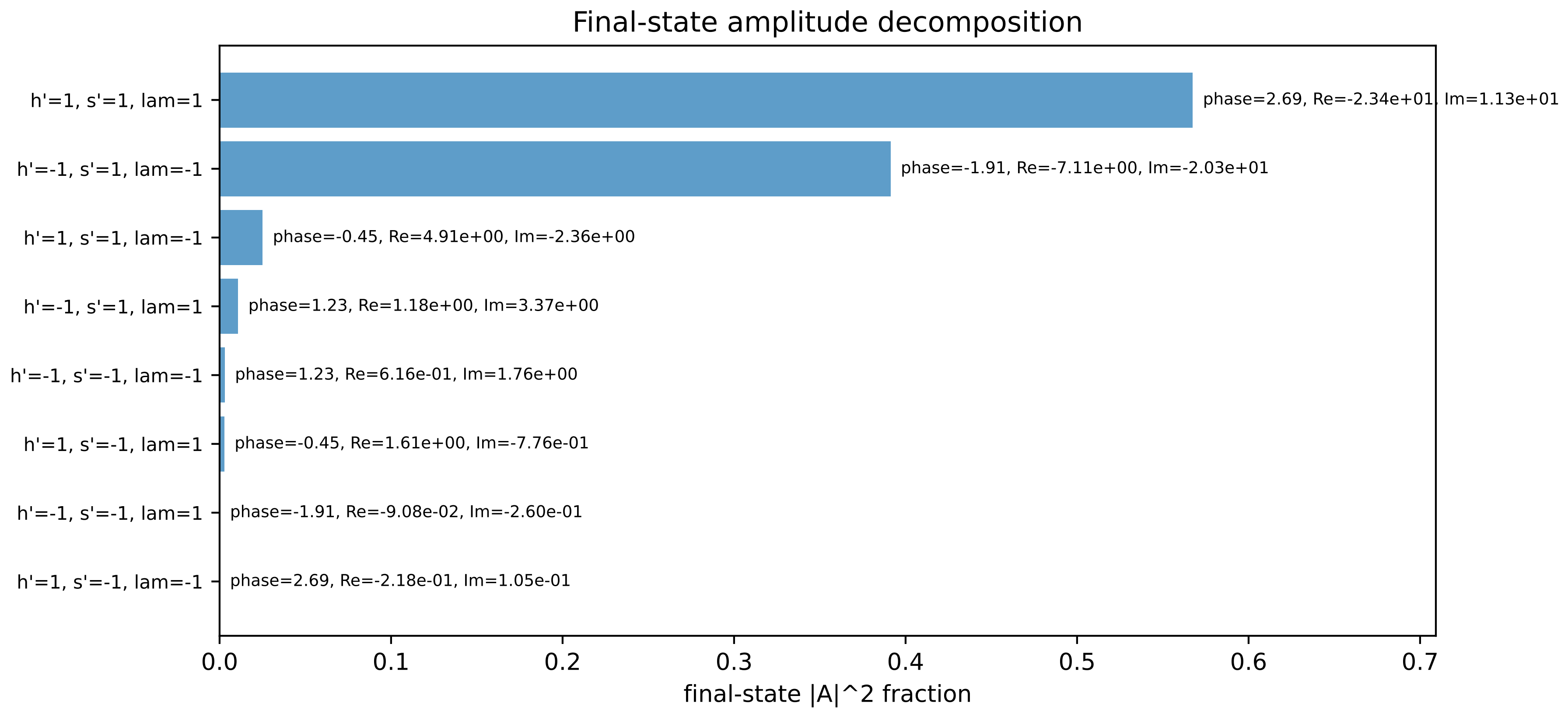


Tx characteristic high-C13 configuration

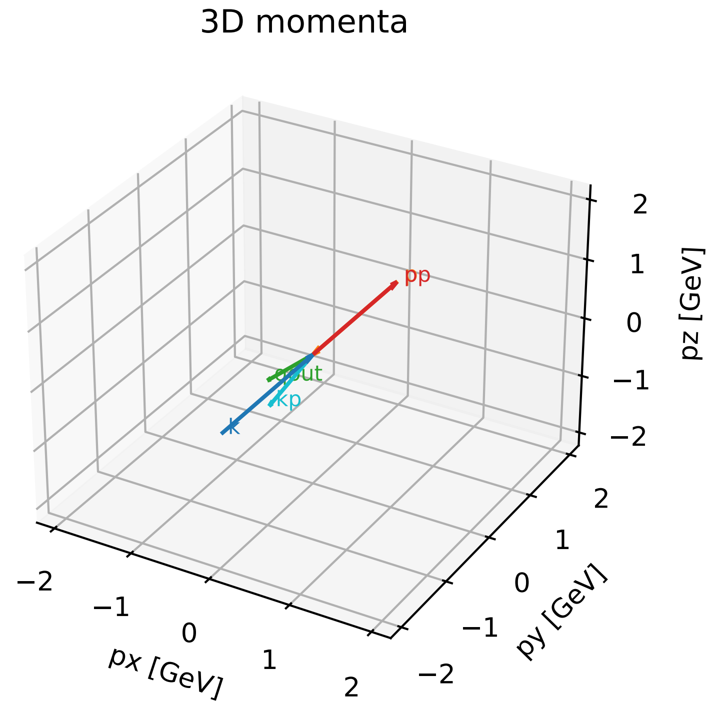



```
Tx_cluster_2  C13=0.914945
kinematic point: mid_s_high_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)

s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=2.14466
theta_in=1.57079, phi_e_in=5.49779, phi_p_in=2.35619
qOut=1.75, phi_gamma=4.90874
Q2=2.16338, xB=0.122099, t=-2.79555, W2=16.4347, y=0.85861
theta(e',gamma)=49.8104 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.2244 px=  1.5729 py= -1.5729 pz= -0.0000
p     E=  2.4141 px= -1.5729 py=  1.5729 pz=  0.0000
kp    E=  0.5477 px= -0.3414 py= -0.4283 pz=  0.0000
pp    E=  2.3408 px=  0.0000 py=  2.1447 pz=  0.0000
qout  E=  1.7500 px=  0.3414 py= -1.7164 pz=  0.0000
```





Ty characteristic high-C13 configuration




```
Ty_cluster_0  C13=0.928651
kinematic point: low_s_high_theta_in_low_Egamma
incoming state: (A[hIn=+1,sIn=1] + i A[hIn=-1,sIn=1]) / sqrt(2)

s=16.7824, sqrt(s)=4.09663, pIn=1.94093, pOut=1.92864
theta_in=1.57079, phi_e_in=4.71239, phi_p_in=1.5708
qOut=0.75, phi_gamma=4.51604
Q2=0.0346994, xB=0.00569867, t=-2.87478e-05, W2=6.93417, y=0.382897
theta(e',gamma)=18.242 deg

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p     E=  2.1557 px=  0.0000 py=  1.9409 pz=  0.0000
kp    E=  1.2020 px=  0.1463 py= -1.1930 pz=  0.0000
pp    E=  2.1446 px=  0.0000 py=  1.9286 pz=  0.0000
qout  E=  0.7500 px= -0.1463 py= -0.7356 pz=  0.0000
```
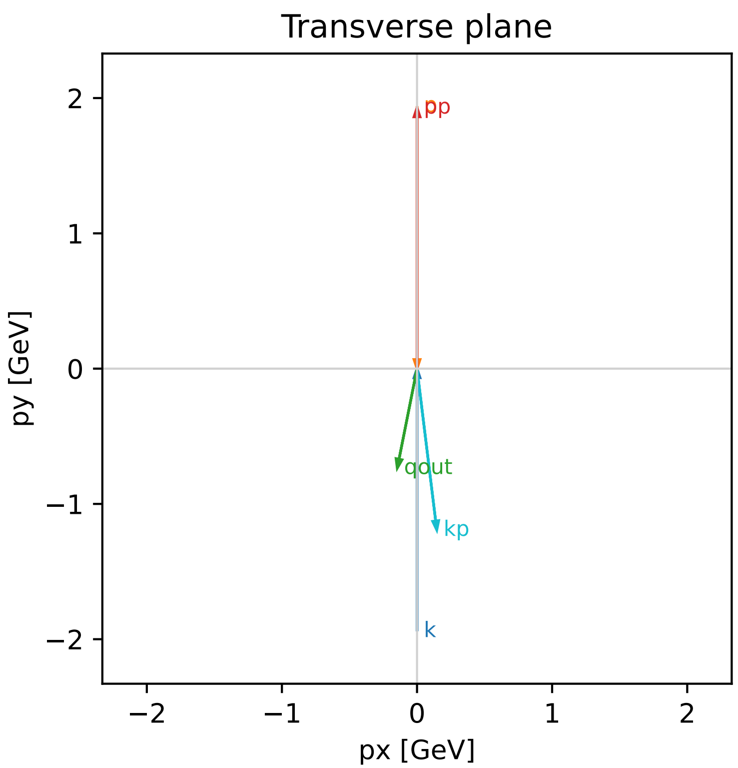
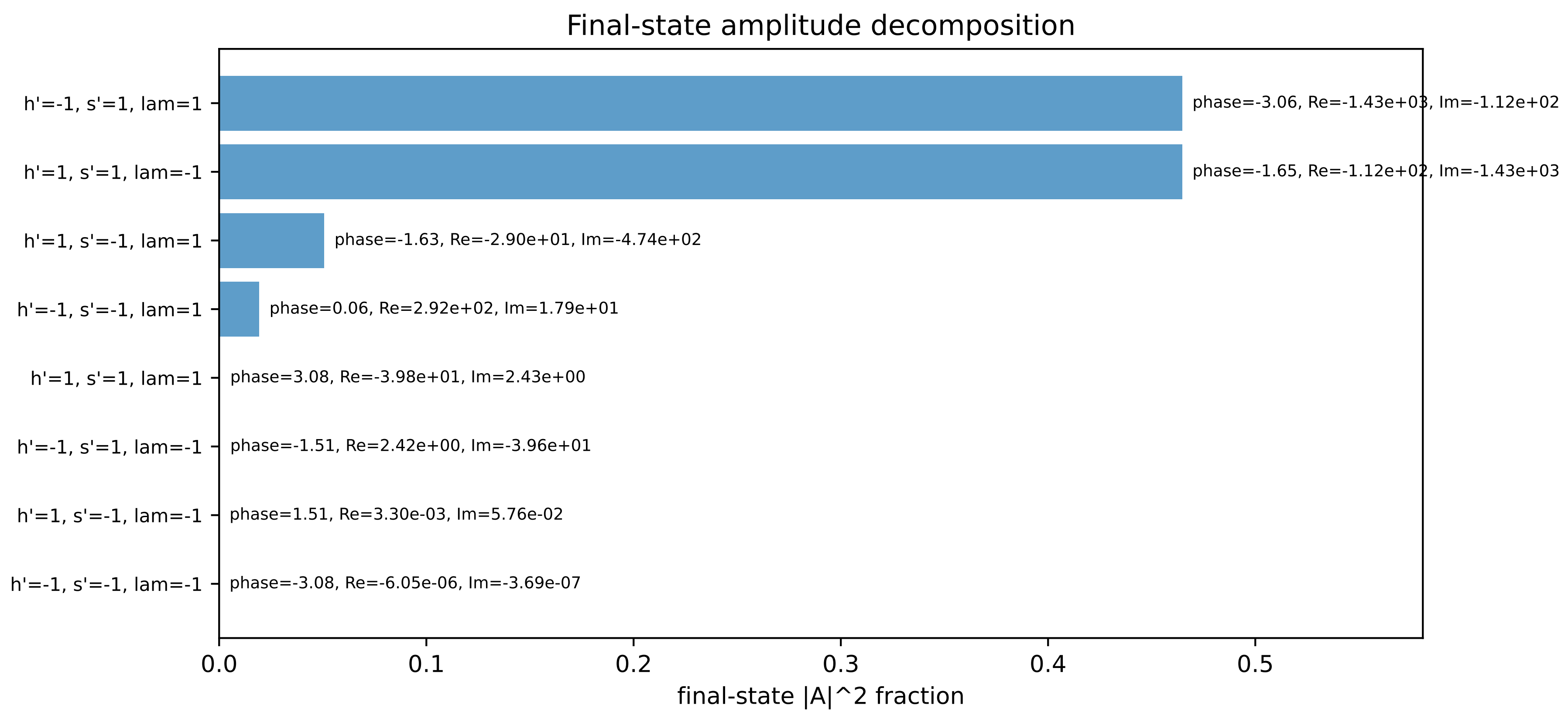
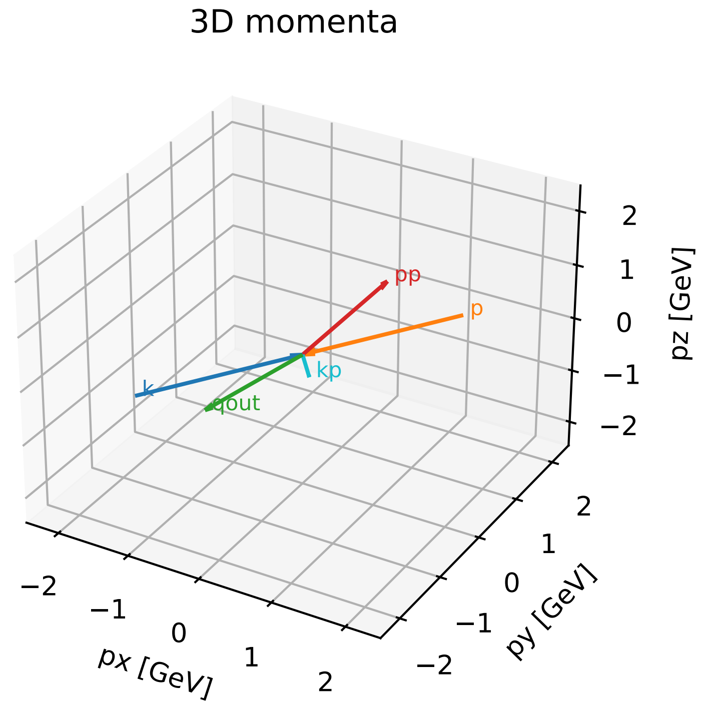
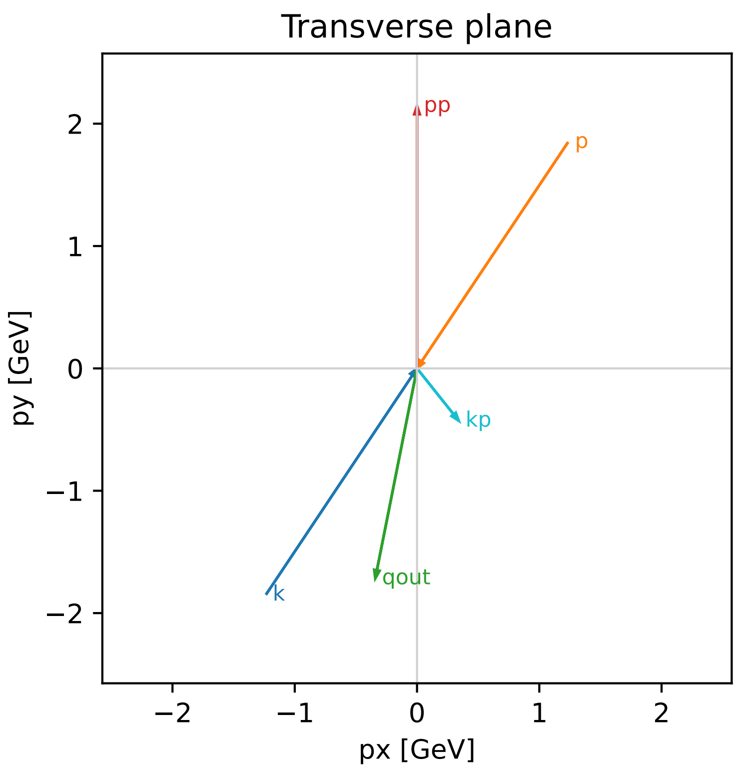



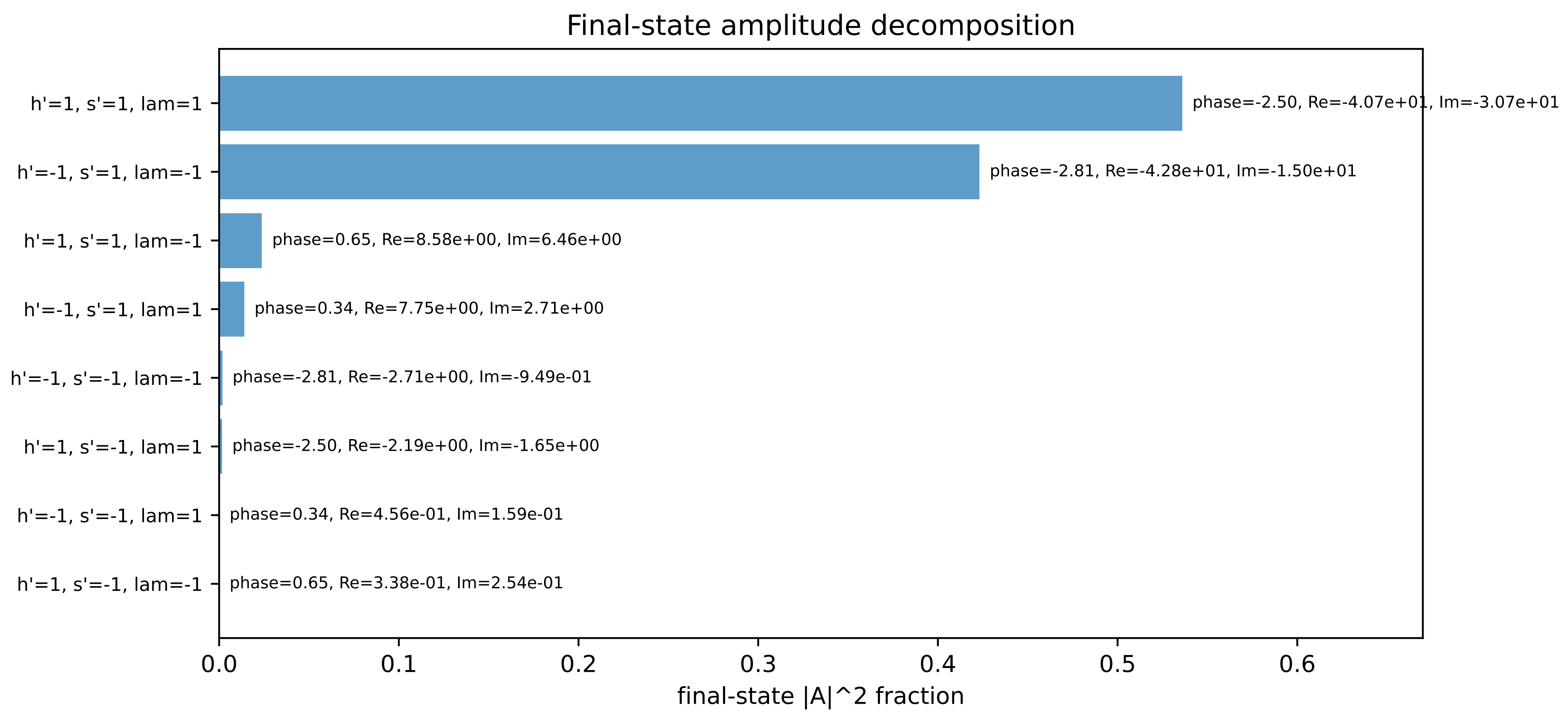


Ty characteristic high-C13 configuration



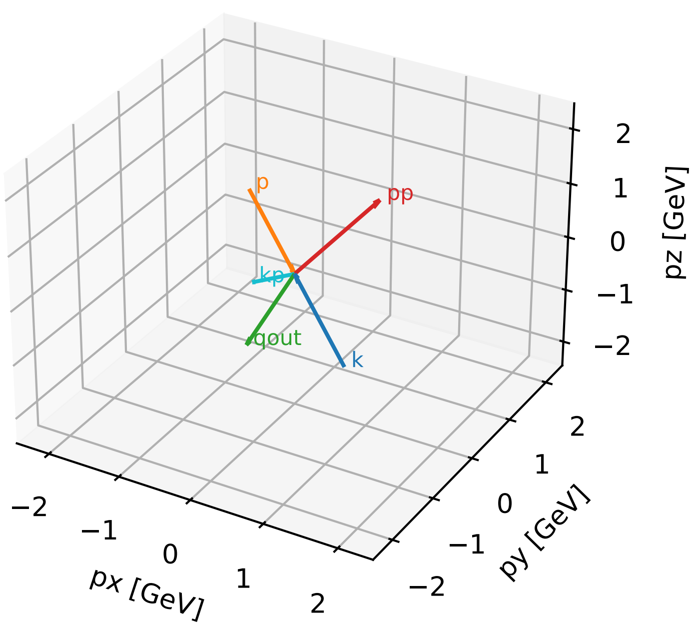

```
Ty_cluster_1  C13=0.919126
kinematic point: mid_s_high_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + i A[hIn=-1,sIn=1]) / sqrt(2)

s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=2.14466
theta_in=1.57079, phi_e_in=4.12334, phi_p_in=0.981748
qOut=1.75, phi_gamma=4.51604
Q2=1.69626, xB=0.0983273, t=-1.60898, W2=16.4347, y=0.835974
theta(e',gamma)=49.8104 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.2244 px= -1.2358 py= -1.8495 pz= -0.0000
p     E=  2.4141 px=  1.2358 py=  1.8495 pz=  0.0000
kp    E=  0.5477 px=  0.3414 py= -0.4283 pz=  0.0000
pp    E=  2.3408 px=  0.0000 py=  2.1447 pz=  0.0000
qout  E=  1.7500 px= -0.3414 py= -1.7164 pz=  0.0000
```





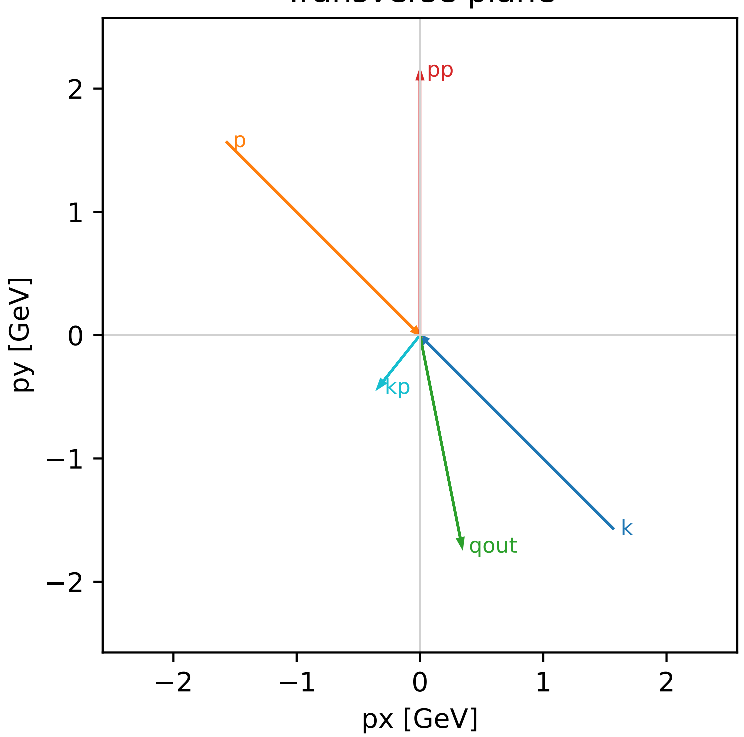
Ty characteristic high-C13 configuration

3D momenta




```
Ty_cluster_2  C13=0.914945
kinematic point: mid_s_high_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + i A[hIn=-1,sIn=1]) / sqrt(2)

s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=2.14466
theta_in=1.57079, phi_e_in=5.49779, phi_p_in=2.35619
qOut=1.75, phi_gamma=4.90874
Q2=2.16338, xB=0.122099, t=-2.79555, W2=16.4347, y=0.85861
theta(e',gamma)=49.8104 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.2244 px=  1.5729 py= -1.5729 pz= -0.0000
p     E=  2.4141 px= -1.5729 py=  1.5729 pz=  0.0000
kp    E=  0.5477 px= -0.3414 py= -0.4283 pz=  0.0000
pp    E=  2.3408 px=  0.0000 py=  2.1447 pz=  0.0000
qout  E=  1.7500 px=  0.3414 py= -1.7164 pz=  0.0000
```

Transverse plane



Final-state amplitude decomposition